

Oct 27, 2024 (Due: 08:00 Nov 3, 2025)

1. Let $(\hat{\lambda}, \hat{x})$ be an approximate eigenpair of $A \in \mathbb{C}^{n \times n}$, and $r = A\hat{x} - \hat{x}\hat{\lambda}$ be the residual. Suppose that $\|\hat{x}\|_2 = 1$. Show that there exists $E \in \mathbb{C}^{n \times n}$ such that

$$(A + E)\hat{x} = \hat{x}\hat{\lambda}, \quad \|E\|_2 \leq \|r\|_2.$$

2. Let $A_0 \in \mathbb{C}^{n \times n}$, $\mu_0, \mu_1, \dots, \mu_m \in \mathbb{C}$. Define $A_1, A_2, \dots, A_{m+1}$ by

$$A_k - \mu_k I = Q_k R_k, \quad A_{k+1} = R_k Q_k + \mu_k I,$$

for $k \in \{0, 1, \dots, m\}$, where Q_k 's are unitary matrices. Show that

$$(A_0 - \mu_0 I)(A_0 - \mu_1 I) \cdots (A_0 - \mu_m I) = (Q_0 Q_1 \cdots Q_m)(R_m \cdots R_1 R_0).$$

3. Let A be the 200×200 Hilbert matrix (the (i, j) -entry of A is $a_{i,j} = (i+j-1)^{-1}$). Compute the eigenvalue of A that is closest to 2 using inverse iteration (INVIT) and Rayleigh quotient iteration (RQI). Visualize the convergence history and report the execution time (preferably with detailed profiling).

4. Write a program to compute all eigenvectors of an upper triangular matrix with distinct diagonal entries. Compare your result with the one produced by a math library (e.g., `eig` in MATLAB/Octave).

5. Implement the following algorithms for Hessenberg reduction:

(a) using Householder reflections;

(b) using Arnoldi process based on modified Gram–Schmidt orthogonalization.

Randomly generate a few matrices and compute the corresponding Hessenberg decomposition $A = QHQ^*$. Check the accuracy in terms of $\|Q^*AQ - H\|_F$ and $\|Q^*Q - I\|_F$ for your Hessenberg reduction implementations. What do you observe?

(optional) Perturb the matrix A a little bit. How do Q and H change accordingly?

6. (H) Let

$$A = \begin{bmatrix} \alpha & \gamma \\ 0 & \beta \end{bmatrix} \in \mathbb{R}^{2 \times 2}.$$

Design an algorithm to compute a unitary matrix $Q \in \mathbb{C}^{2 \times 2}$ such that

$$Q^*AQ = \begin{bmatrix} \beta & \tilde{\gamma} \\ 0 & \alpha \end{bmatrix}.$$

Is it possible to impose $\tilde{\gamma} = \gamma$?

7. (optional) Implement the naive QR algorithm and visualize the *componentwise* convergence on a small example.

8. (H, optional) Write a subprogram to perform diagonal swapping of the real Schur form

$$A = \begin{bmatrix} A_{1,1} & A_{1,2} \\ & A_{2,2} \end{bmatrix},$$

where $A_{1,1}$ and $A_{2,2}$ are either 1×1 or 2×2 .